

sequential, whereas the 2421 and 5211 codes are not sequential.

Some features of weighted codes

Except 8421 and 74-2-1 codes, for all other weighted codes some of the decimal number may be coded in more than one forms. For example, decimal 4 in 2421 code may be represented as 1010 or 0100. That's why 2421 and 4221 are known as self-complementing codes.

A property known as self-complementing is applied to select a code out of different options. The property is stated as follows: If 'D' is the given decimal number, you may code it in any option, but then decimal number 9-D must be coded taking the option such that the code is bit-wise complement of the code of D. For example, if decimal 4 in 2421 is coded as 0100, then 9-4=5 must be coded as 1011. See that 1011 is bit-wise complement of 0100.

We have shown the code in Table II applying the self-complementing property. A necessary condition of self-complementing property is that the sum of weights in the code is 9. The codes 8421 and 74-2-1 are not self-complementing.

Non-weighted code

In many cases, there may be some requirement for non-weighted codes (see Table III), particularly from design point of view. Non-weighted codes are codes that are not positionally weighted. That is, each position within the binary number is not assigned a fixed value. The examples of a few non-weighted codes are excess-3 code, 1-to-2 code and BCDP (BCD with parity).

Excess-3 is a non-weighted code used to express decimal numbers. The code derives its name from the fact that each binary code is the corresponding 8421 code plus 0011(3).

Some features of non-weighted code

Excess-3 code is just BCD with plus 3 weight. If 'd' is the decimal number, d+3 is coded in 8421 to get the excess-3

TABLE II
Weighted Codes

Decimal digit	2421 code	74-2-1 code	4221 code
	Weight 2 4 2 1	Weight 7 4 2 -1	Weight 4 2 2 1
0	0 0 0 0	0 0 0 0	0 0 0 0
1	0 0 0 1	0 1 1 1	0 0 0 1
2	0 0 1 0	0 1 1 0	0 0 1 0
3	0 0 1 1	0 1 0 1	0 0 1 1
4	0 1 0 0	0 1 0 0	0 1 1 0
5	1 0 1 1	1 0 1 0	1 0 0 1
6	1 1 0 0	1 0 0 1	1 1 0 0
7	1 1 0 1	1 0 0 0	1 1 0 1
8	1 1 1 0	1 1 1 1	1 1 1 0
9	1 1 1 1	1 1 1 0	1 1 1 1

TABLE III
Non-weighted Codes

Decimal digit	Excess-3 code (XS-3 code)	1-to-2 code	BCDP (with odd parity)
0	0 0 1 1	0 0 0 1	0 0 0 0 1
1	0 1 0 0	0 0 1 0	0 0 0 1 0
2	0 1 0 1	0 0 1 1	0 0 1 0 0
3	0 1 1 0	0 1 0 0	0 0 1 1 1
4	0 1 1 1	0 1 0 1	0 1 0 0 0
5	1 0 0 0	0 1 1 0	0 1 0 1 1
6	1 0 0 1	1 0 0 0	0 1 1 0 1
7	1 0 1 0	1 0 0 1	0 1 1 1 0
8	1 0 1 1	1 0 1 0	1 0 0 0 0
9	1 1 0 1	1 1 0 0	1 0 0 1 1

code. This code is the most important non-weighted code in digital logic design. 1-to-2 code has a unique feature. No code has less than one and greater than two 1's. BCDP (with odd parity) is designed so as to make any code having odd numbers of 1's. This is done with selection of parity bit, which is the right-most bit of the code.

Gray code/binary reflected code

Gray code is a very important non-weighted code. An N-bit gray code is a sequence of all the N-bit binary numbers, ordered in such a way that each binary number differs from its predecessor and successor by exactly 1 bit (and the first and last differ by one bit also). For example, one-bit gray code

0, 1 and three-bit gray code 000, 001, 011, 010, 110, 111, 101, 100 (differs from 000 by 1 bit). The sequence 00, 01, 10, 11 is not a two-bit gray code as the first and second elements differ by two bits.

Table IV shows a 4-bit code in comparison with a binary number. In the column of decimal number, it is seen that from 00 to 09 if you move from number N to number N+1, there is a change in

only one digit. But when you move from 09 to 10, there are changes in two digit positions. Look at the column of reflected decimal number: as you move from N to N+1, always there is change in one digit position.

In binary representation as we move from N to N+1, there may be many changes in bits. For example, as you go from 3 to 4, there are as many as three changes in bit position. But, look at the reflected binary or gray code: as you move from any N to N+1, always there is only one change in bit position.

This is the property of gray code or reflected binary code. Gray code is also known as a variable weighted code and is cyclic.

The gray code is called 'reflected binary,' because the first eight values compare with those of the last eight values, but in reverse order. The gray code belongs to a class of codes called 'minimum change codes,' in which only one bit in the code changes when moving from one code to the next. The gray code is a reflective digital code which has a special property that any two subsequent numbers' codes differ by only one bit. It is also called 'unit-distance code.'

Gray code has an important application in digital control systems. Let